

Calibration and Uncertainty of Molecular Property Prediction Models under Distribution Shift

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1 Motivation

Molecular property prediction models are a core component of modern drug discovery pipelines, where they are used to prioritize candidate compounds before costly experimental validation. In practical settings, these models are frequently applied to molecules that differ structurally from those seen during training, leading to distribution shift.

While predictive performance under such shifts has been studied extensively, the reliability of uncertainty estimates and calibration of molecular prediction models remains less well understood. Poorly calibrated models may produce overconfident predictions, which is particularly problematic in safety-critical applications such as bioactivity or toxicity prediction.

This project aims to study how distribution shift affects uncertainty estimation and calibration in molecular property prediction models, and to evaluate whether commonly used uncertainty-aware techniques improve reliability under realistic evaluation settings.

2 Research Question

How does distribution shift induced by scaffold-based splits affect calibration and epistemic uncertainty in graph neural networks for molecular property prediction, and which uncertainty-aware methods mitigate this effect?

To address this question, we consider the following sub-questions:

- How does calibration differ between standard random splits and scaffold-based splits?
- Do uncertainty-aware methods such as ensembles and Monte Carlo dropout improve calibration under distribution shift?
- Is improved calibration associated with better alignment between predictive uncertainty and model error?

3 Problem Setup

We consider the task of *molecular property prediction* (also known as QSAR), where a machine learning model predicts a biochemical property of a molecule given its structure.

- **Input:** molecular graphs derived from chemical structures
- **Output:** a molecular property, formulated as either a binary classification or regression task

Graph Neural Networks (GNNs) are used as the primary modeling approach, as they naturally represent molecular structure and are widely adopted in cheminformatics [3, 4].

4 Methodology

4.1 Models

We compare the following models:

- A standard Graph Neural Network (baseline)
- An ensemble of independently trained GNNs [1]
- A GNN with Monte Carlo Dropout applied at inference time [2]

All models share the same base architecture to isolate the effect of uncertainty estimation methods.

4.2 Distribution Shift

To simulate realistic distribution shift, we evaluate models under two data splitting strategies:

- **Random split**, corresponding to an i.i.d. evaluation setting
- **Scaffold-based split**, where test molecules contain chemical scaffolds not observed during training

Scaffold-based splitting is commonly used in cheminformatics to assess generalization to structurally novel compounds [4]

5 Evaluation Metrics

- Predictive performance is evaluated using: ROC-AUC for classification tasks, or RMSE for regression tasks
- Model calibration is assessed using: Expected Calibration Error (ECE), Reliability diagrams
- Uncertainty Analysis: The relationship between predictive variance and prediction error, Correlation between uncertainty estimates and model errors

6 Experimental Plan

The experimental workflow consists of the following steps:

1. Train a baseline GNN under both random and scaffold-based splits
2. Evaluate predictive performance and calibration in both settings
3. Introduce uncertainty-aware methods (ensembles, MC dropout)
4. Compare calibration and uncertainty quality across methods
5. Analyze the effect of distribution shift on reliability and uncertainty–error alignment

7 Expected Contributions

This project provides an empirical analysis of calibration and uncertainty estimation in molecular property prediction under realistic distribution shifts. By systematically comparing uncertainty-aware methods under scaffold-based evaluation, the project aims to clarify when and how these techniques improve reliability beyond standard predictive performance metrics.

8 Reproducibility

All experiments will be conducted using publicly available datasets and implemented in a reproducible pipeline with fixed random seeds and documented hyperparameters.

References

- [1] Yarin Gal and Zoubin Ghahramani. Dropout as a bayesian approximation: Representing model uncertainty in deep learning. *International Conference on Machine Learning*, 2016.
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- [3] Oliver Wieder, Stefan Kohlbacher, et al. A compact review of molecular property prediction with graph neural networks. *Drug Discovery Today: Technologies*, 37:1–12, 2020.
- [4] Zhenqin Wu, Bharath Ramsundar, Evan N Feinberg, Joseph Gomes, Caleb Geniesse, Aneesh Pappu, Karl Leswing, and Vijay Pande. Moleculenet: a benchmark for molecular machine learning. *Chemical Science*, 9(2):513–530, 2018.